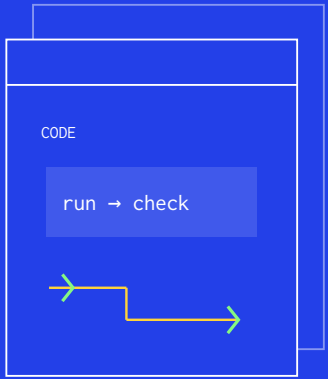
WEEK 2 / LAB 4 / TEAM WORK

MLOps: Machine Learning Operations

Serve the model. Fail honestly.

Expose the evaluated bike-demand model through a validated API and verify it in the container. Needs the Lab 3 work merged into your integration branch (main or develop).



Done means checked behaviour, not a running process

Three outcomes you can demonstrate from the team repository.

1 **Serve**

Load the saved, evaluated model at startup and return its actual version. Never train inside the service.

2 **Validate**

Reject missing fields, wrong types and out-of-range values before inference.

3 **Verify**

Automated tests cover normal, invalid and missing-model cases, and they run inside the container.

Who does what

The split is yours; every member owns something and reviews something.

INDIVIDUAL

One PR each

One area each: validation, load + health/predict, tests, or container + README.

TEAM

One service

Agree contract, merge order and the final container check.

REVIEW

Another member

Break a PR you did not write: one bad request, one missing model.

Individual report reports/week-2.md covers Labs 3–4 in your personal repository.

Add these to the team repository

Paths are the contract; contents are your design.

REQUIRED

Reviewed PRs into your integration branch. Declare new dependencies in pyproject.toml and keep uv.lock in step.

TEAM REPOSITORY PATHS

```
src/bike_demand/api.py
tests/test_api.py
Dockerfile
pyproject.toml uv.lock
README.md
your team's own report
```

README: how to run and test the service. Report: your team chooses its format; add a Lab 4 section.

Two endpoints, three outcomes

Both share one model-availability state.

GET /health

200 or 503

200 only when a usable model is loaded; otherwise 503.

POST /predict

200 + version

JSON with a finite non-negative prediction and the loaded model_version.

OTHERWISE

4xx or 503

Invalid input gets a 4xx; valid input without a model gets 503. Never a fake prediction.

model_version comes from the artifact's own metadata, not a hard-coded string.

Input contract: twelve fields

One flat JSON object with the same fields used in training.

RANGES

Weather values are normalised, not Celsius or percentages. cnt, casual and registered are never accepted as inputs.

REQUEST FIELDS

```
season 1-4 yr 0-1
mnth 1-12 hr 0-23
holiday 0-1 weekday 0-6
workingday 0-1 weathersit 1-4
temp atemp hum windspeed
finite numbers in 0..1
```

Decide and document how extra fields are handled.

Example: call a service and read the answer

A toy /score endpoint; the shape of exchange is what matters.

ILLUSTRATIVE EXAMPLE

A valid request returns 200 and JSON. A bad value is rejected with 422 before any model runs.

HTTP EXCHANGE EXAMPLE

```
curl -X POST localhost:8000/score \
-H "Content-Type: application/json" \
-d '{"size": 4}'

200 {"score": 0.82, "model_version": "m-01"}
422 {"detail": [{"loc": ["body", "size"]}]}
```

Different project on purpose: read it, then adapt it to yours; do not paste it.

Tools you will meet

Lecture 4 covered API concepts; these are this lab's concrete tools.

BUILD

FastAPI + Pydantic

Routes, request models with field constraints, and an OpenAPI page at /docs.

TEST

pytest + TestClient

Call the app without a server; start each case in the state it tests.

RUN

uvicorn + docker

Listen on 0.0.0.0 and publish a port to reach it.

Read the official documentation for each; choosing how to use them is the lab.

Example: a request model and an endpoint

FastAPI reads the type hints; Pydantic enforces the constraints.

ILLUSTRATIVE EXAMPLE

FastAPI reads the type hints; Pydantic rejects a value outside the range before your code runs. Open /docs to see the generated contract.

APP EXAMPLE (PART 1)

```
from fastapi import FastAPI
from pydantic import BaseModel, Field
class Item(BaseModel): size: int = Field(ge=0, le=10)
app = FastAPI()
```

Shown as one-liners to fit the slide; in a real file the class body is indented on its own line.

Example: answer 503 when the model is missing

Fail with an honest status instead of guessing a number.

ILLUSTRATIVE EXAMPLE

Raise an HTTPException with 503. How you hold and check the model state is your design choice.

APP EXAMPLE (PART 2)

```
from fastapi import HTTPException
state = {"model": None}
def stop(): raise HTTPException(503, "no model")
@app.post("/score")
def score(i: Item): return state["model"] or stop()
```

A health route reads the same state and returns 200 or 503. Different project on purpose: adapt, do not paste.

Example: test the service without a server

TestClient calls the app in-process; no port is needed.

ILLUSTRATIVE EXAMPLE

One case per behaviour. Put these lines inside a function named test_something; pytest collects it.

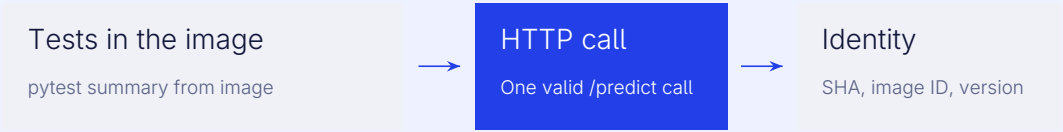
PYTEST EXAMPLE

```
from fastapi.testclient import TestClient
from app import app
client = TestClient(app)
bad = {"size": 40}
r = client.post("/score", json=bad)
assert r.status_code == 422
```

Needs httpx; add it with uv add --dev httpx so the lock stays in step.

Prove it where it will run

Evidence comes from the built image, not from your laptop environment.



`A passing local pytest does not show the container tests ran.`

Keep raw logs your report can link to; at most three screenshots.

Example: run the service from its image

Same image, two different commands.

ILLUSTRATIVE EXAMPLE

Listen on 0.0.0.0 inside the container and publish the port to your own machine only.

TERMINAL EXAMPLE

```
docker build -t svc:dev .
docker run --rm -p 127.0.0.1:8000:8000 svc:dev
curl localhost:8000/health
docker run --rm svc:dev pytest -q
```

The start command (for example uvicorn app:app --host 0.0.0.0) lives in your Dockerfile CMD.

m2 and your individual report

One team checkpoint for Labs 3–4; one personal report.

TEAM

Annotated tag m2

On your integration branch after review and checks; submit the link and a note.

PERSONAL

reports/week-2.md

Your contribution, one decision, one blocker, assistance used.

RULES

Honest and on time

Personal main, no PR or CI. Sunday 23:59. 10% of course.

Incomplete work is reported as incomplete; never invent a result or a passing check.

Example: tag the checkpoint

An annotated tag carries a message and points at a reviewed commit on your integration branch.

ILLUSTRATIVE EXAMPLE

Tag only after review and current checks, then push the tag so the link works.

GIT EXAMPLE

```
git switch develop
git pull
git tag -a m2 -m "Week 2 checkpoint"
git push origin m2
```

Use main instead of develop if that is your branch. Submit the tag link with a short note.

EXIT CHECK

A usable service tells the truth when it fails.

Next, Lab 5: local deployment, structured logs and serving-version checks.